

Arrays have some disadvantages:

1. Fixed size (we can't increase/decrease)
2. Size should be known beforehand.

ArrayList has advantages

1. Dynamic size
2. Don't need to know size beforehand

1. Shopping cart
2. New Tabs of browser
3. Song Playlist

Syntax

```
ArrayList <Type> arr = new ArrayList<Type>();
```

↳ Type can't be primitive

↓
int, long, float,
double, char

↳ we can use wrapper classes

Primitive Data Type

Wrapper class

int

Integer

long

Long

⋮

Wrapper classes are built around primitive data type. Contain additional methods for user to use

Basic Operations

1. Insert element $\rightarrow$ add()

arr.add(5)

element is added at end of arraylist

2. Size

arr.size()

3. get

arr.get(i) access element at ith idx

4. remove

`arr.remove(i)` removes *i*th idx element from array list

5. set

$\begin{matrix} 0 & 1 & 2 & 3 \\ < 1, & 5, & 4, & 3 > \end{matrix}$
↓

`arr.set(1, 8)` $< 1, 8, 4, 3 >$

updates the value of a particular idx with new element

`arr.set(idx, new-elm)`

10:19

x_0	1	2	6	7
x_1	8	9	10	20
x_2	30	1	5	7

3x4

→ [[1, 2, 6, 7],

→ [8, 9, 10, 20],

→ [30, 1, 5, 7]]

↓

[x₀,

x₁,

x₂]

2D Array → Array of arrays

→ Array of rows

A = < 1, 4, 6, 8 >

B = < 2, 5, 7, 9 >

Ⓐ = [[1, 4],
[2, 5, 7, 9]]

ArrayList<Integer> row0 = A.get(0)

int row0C = new int[A.get(0).size()]

Array ⊇ ArrayList

int x = 2

int y = 4

if (x == y)